

William Andrian Dharma T

Artificial Intelligence Laboratory Assistant @ STEI ITB

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Summary

Artificial Intelligence Laboratory Assistant at STEI ITB as a third year undergraduate student majoring in Informatics Engineering at Institut Teknologi Bandung (ITB). Primarily interested in Data Science, Machine Learning, and Artificial Intelligence. Has additional experience with full-stack Web Development and deployment.

Skills

- **Machine Learning Tasks:** Tabular Data, Image Recognition, Image Segmentation, Natural Language Processing, Sentiment Analysis, Audio Recognition, Time series analysis and forecasting
- **Machine Learning Frameworks:** Sklearn, Pytorch, Transformers, Tensorflow
- **Artificial Intelligence Tasks:** Large Language Models (LLMs), Retrieval Augmented Generation (RAG)
- **Artificial Intelligence Tools:** ChromaDB, LangChain
- **Web Technologies:** React JS, Flask, FastAPI, Gin, Coolify CI/CD, PostgreSQL, Docker
- **Programming Languages:** Python, Go, SQL, Javascript, Typescript, Java, C++
- **Languages:** Indonesian (Mother Tongue), English (Fluent, IELTS band 8), Japanese (JLPT N2)

Education

Institut Teknologi Bandung (2023-2027 expected)

Information Engineering

Latest GPA : 3.63

Experience

Artificial Intelligence Laboratory Assistant

STEI ITB

September 2025 - Present

- Create Prolog-based course tasks for the Computational Logic course
- Grade quizzes for the Computational Logic course

Data Science Competition and Community Staff

HMIF ITB

October 2024 - Present

- Lead solution notebook reveals for internal data science playgrounds

Data Scientist

Tanto Intim Line

July 2025 - January 2026

- Built an automatic data processor and report generator desktop app in Java
- Conducted data analysis on company car fuel spending, identifying falsified reimbursement reports for personal gain.
- Conducted data analysis for container ship fuel consumption, comparing different vessels and routes

Projects

ML Models From Scratch

<https://github.com/kirisame-ame/ML-Models-From-Scratch>

Supervised, Unsupervised, and Reinforcement Learning algorithms made from scratch with only math libraries in python

Songwiz

<https://songwiz.kirisame.jp.net/>

Song query website by audio or album image similarity. Audio similarity is done on MFCC Chroma with euclidean distance. MIDI similarity uses FTB, RTB, and ATB with cosine similarity. Image similarity is done by PCA with euclidean distance.

Retrieval Augmented Generation (RAG) Chatbot

<https://chat.kirisame.jp.net/> (Active in development)

A Retrieval Augmented Generation chatbot on personal files created with Langchain, ChromaDB, Gemini Embedding, GPT-OSS via Openrouter, run on FastAPI, displayed with a react app, dockerized and deployed on a Coolify instance on a VPS. Also experimented with prompt engineering to output in a desired manner and defend against prompt injection.

Twitter Sentiment Analysis

https://github.com/kirisame-ame/text_classification

Natural Language Processing (NLP) notebook for sentiment analysis on an Indonesian twitter dataset. Preprocessed the text with regex, nltk stopwords, and tokenized with Count vectorizer and tfidf before being classified with a catboost model. A separate notebook using the BERT transformer is also available and resulted in greatly increased model accuracy.

Research

Data Science

[Using Graph Theory to Analyze Passing Maps of Premier League Teams and Identify Key Players](#)

A paper on football about utilizing graph theory to analyze passing maps of premier league teams, and identify standout players in their passing network and the team’s overall passing structure. Using metrics such as closeness centrality, betweenness centrality, eigenvector centrality, transitivity, and constructing a minimum spanning arborescence, we are able to calculate a player’s impact on their team’s passing network and the team’s passing tendencies. In this paper we demonstrated this on a match between Liverpool and Manchester United on 5 January 2022, and were able to point out several key players such as Andy Robertson, Trent Alexander-Arnold, and Alexis Mac Allister for Liverpool, and Bruno Fernandes as the main player for Manchester United. Our results were in accordance to actual analyses by experts and pundits, and also by the fact that Bruno Fernandes won man of the match in that game.

Awards and Certifications

1st Place - Data Science ARA 7.0 by Institut Teknologi Sepuluh Nopember (ITS)

Pothole image semantic segmentation competition using Machine Learning models. For this computer vision task specifically on semantic segmentation, our team experimented a lot with preprocessing pipelines and model selection. We tried several architectures from CNNs to transformers and ended up with the Encoder-only Mask Transformer (EoMT) as our final model. The final stage had us demonstrate our understanding of both the problem at hand and our solution by presentation and a QnA session with the judges.

Japanese Language Proficiency Test (JLPT) N2

Global Consumer Intelligence Course by Matsuo-Iwasawa Lab U-Tokyo

A course hosted by The University of Tokyo’s Matsuo-Iwasawa Lab about Data Science and Artificial Intelligence, and how it is applicable in business.

International English Language Testing System (IELTS) Band 8.0